SINKGRAB · STEM CHECK

SinkGrab STEM check

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DOCUMENT CONTROL

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Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/sinkgrab		
Source	docs/01b-stem-check.md @ 780e4b8		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-25	Amish Chadha	Amish's decisions of 2026-10-25 ("i approve of your recommendations", SKG-DEC-001 O2, O4, O5): rows S4 to S7, S10, S10a, S15, S16, S18, S21, E2, B4 and OB2 restated for the bentonite annulus, the crank slip clutch and load indicator with proof at 1.25 x the rated load, and four people at the T-bar; S6, S7, S21, E2 and B4 now plausible, to confirm at TRL 4; 33 within, 13 plausible, 2 failing (option B); verdict open
0.2	2026-10-25	Amish Chadha	Independent review recorded: seven rows now fail with a fix proposed to Amish (S6, S7, S21, E2, B4, OB5, OB6), six rows downgraded to plausible or added (S2, S8, S10a, S14a, S22), K22 decided under the standing instruction (gas read at depth); verdict open
0.1	2026-10-24	Amish Chadha	First STEM check (kit 1.9.1 upgrade of a TRL 3 design): 46 claims for options A and B on the detailed model; five failed details fixed under Amish's standing instruction of 2026-10-23

	(SKG-DEC-001 K1 to K5); independent review; verdict open

SinkGrab STEM check

Verdict: open (two rows fail with a fix proposed, both for option B; thirteen to confirm by trial). SinkGrab's mechanism, digging a flooded caisson well from the surface with a rope grab and an under-curb scraper on a hand capstan, sits within known science. The independent review of 2026-10-25 found two things it relied on did not hold on paper: the rings sinking 3 m without de-watering (S21) and a release window for the shear devices between the working pull and the HatchSide tripod's rating (S6, S7, E2), and the scraper's turning force was over sustained limits (B4). Amish decided the fixes on 2026-10-25 ("i approve of your recommendations", SKG-DEC-001 O2, O4, O5): a bentonite slurry in the overcut annulus is part of the sinking method; a friction slip clutch on the crank shaft and a load indicator at the dead end replace both shear devices, with proof at 1.25 times the rated load; and four people turn the scraper. Each rests on a figure only a trial can give (the lubricated friction, the clutch's setting tolerance, the toe force), so those rows are now plausible, to confirm at TRL 4. Of 48 claims for both concept options, 33 are within known science, 13 are plausible and can only be confirmed by a trial at TRL 4 (S2, S3, S5, S6, S7, S8, S10a, S14a, S21, S22, E2, B4, OB3), and two fail with a fix proposed for option B only (OB5, OB6), which waits on the concept choice (O1). `project.yaml` records `stem_check: open`. Six claims failed when first checked on the detailed model and each is now fixed in the design under Amish's standing instruction of 2026-10-23 ("when a STEM check row fails, propose and implement the best fix for the design and its constructability"): the shear link's calibrated pin bent across an open gap (since replaced under O4), the brake's weight slackened the band, the assembled grab was lifted by two people, the full spoil tub had no safe way off the doors, the gas detector did not read carbon dioxide (SKG-DEC-001 K1 to K5), and, from the review, the gas was read only at the collar (K22). The scraper arm's eye (S20) was fixed as design for construction (K6). Method: STANDARDS section 20, fix rule. Calculations: `docs/04-calcs/stem_check.py`, results in `docs/04-calcs/stem_results.csv`.

Claims checked

Table 1. Claims, governing science and verdicts. Rows S5, S11, S20, B1, B2 and C2 failed when first checked; the check column says what failed and what fixed it. Rows marked "revised after the independent review" changed verdict on 2026-10-25. "Open" rows are closed only by a trial at TRL 4; "Awaiting Amish" rows by his decision.

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
M1	Closed shells hold 28.3 L; a bite is 21.2 L at the assumed 75 % fill	Mathematics	Units and dimensional analysis	Two quarter cylinders, V = pi R^2 B / 2	claimed 28.3 L, estimate 28.3 L (ratio 1.00)	**Within known science**	SKG-CAL-001 A1
S1	3:1 tackle closes the lips with about 1.06 times the line pull	Physics	Mechanics and statics	Virtual work: line travel / lip travel (frictionless)	claimed 1.06 x line pull, estimate 1.06 x line pull (ratio 1.00)	**Within known science**	SKG-CAL-001 B2
S2	The lips close a full bite in loose saturated sand (force per shell against passive drag at full bite depth)	Physics	Mechanics and statics	Rankine passive resistance 0.5 gamma' d^2 Kp B, phi' 30 deg, gamma' 10 kN/m3, d = shell radius; virtual work through the 3:1 tackle	Drag about 270 N per shell against 349 N per shell with the full grab's submerged weight on the line. Revised after the independent review: closing starts with the grab empty (450 N submerged), sheave friction (three sheaves, 10 to 15 %) is left out, whether 1.06 T acts on each lip or on both is not settled, and denser sand (phi' 40 deg) roughly doubles the drag; the margin lies between about 0.9 and 1.8, so the bite stays open until the test-pit trial	**Plausible, to confirm at TRL 3**	Open: see below
S3	The head's weight opens the shells when the line is slacked on the bottom	Physics	Mechanics and statics	Single-rope grabs open under their own head weight on slack line; resistance is the shells' pins and the sand lying on the lips	head and ballast 14.3 kg; the lips ride up and outward over the surface as they open; confirmed only by a bite trial	**Plausible, to confirm at TRL 3**	Open: see below

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S4	The steady pull of the full grab at the capstan stays below the slip clutch's lowest slip	Physics	Mechanics and statics	Weight of the full grab in air x 1.11 sheave and guide tube friction against the clutch's slip on the third layer at the low end of a +/-5 % setting	1068 N <= 1127 N	**Within known science**	SKG-CAL-001 C4; SKG-DEC-001 O4
S5	The dead-end load indicator shows the highest pull of each lift	Physics	Mechanics and statics	Hooke's law for a stack of disc springs in series (deflection adds, load is shared); a friction tell-tale on the rod keeps the largest travel	Restated for Amish's decision O4 of 2026-10-25: the shear link and its calibrated pin are gone (they were fixed under K1 for true double shear). Fourteen disc springs 18 x 9.2 x 1.0 give about 2.8 mm of rod travel at 1.47 kN on catalogue values to confirm; disc springs have some hysteresis and the tell-tale reads to about one 0.25 kN ring; calibrated on CalRig and checked in the mud at TRL 4	**Plausible, to confirm at TRL 3**	Open: TRL 4 (SKG-DEC-001 O4, decided 2026-10-25)
S6	The crank slip clutch caps the capstan pull at the HatchSide tripod's 150 kg rating	Physics	Mechanics and statics	Friction clutch, T = 2 mu F r_m (two faces, uniform pressure), mu 0.35 for a dry woven facing on ground steel; 4:1 chain; innermost rope layer	Restated for Amish's decision O4 of 2026-10-25 (it replaces the 3 mm shear pin, which ran above shear yield in normal hoisting): a 33.8 N m slip from a 1464 N clamp gives 1472 N at the high end of a +/-5 % setting on the first layer. The facing friction moves with	**Plausible, to confirm at TRL 3**	Open: TRL 4 (SKG-DEC-001 O4, decided 2026-10-25)

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
					moisture and glazing, so the setting rests on CalRig before each campaign and a dry lining inside the guard; in ordinary hoisting it carries about 81% of its slip torque, which a friction face takes without fatigue		
S7	The slip window carries the full grab on every rope layer and stays inside the rating	Physics	Mechanics and statics	One slip torque gives a line pull that falls as the rope builds up on the drum (layer radii 88 to 104 mm); sheave and guide tube friction about 11 %	Lowest slip 1127 N (layer 3, low end) against a steady pull of 1068 N, a margin of 1.06; highest 1472 N against the 1472 N rating. A jerked start or breakout suction (about 160 N) makes it slip for a moment and the pawl holds; nothing releases. At +/-10 % the layer 3 low end falls to about 1019 N, under the steady pull, so the narrow window is confirmed on CalRig (TRL 4)	**Plausible, to confirm at TRL 3**	Open: TRL 4 (SKG-DEC-001 O4, decided 2026-10-25)
S8	Hoisting power per person is sustainable for a working day	Biology	Human physiology and biomechanics	Sustained arm cranking by an adult; CAL assumes 50 W each against about 75 W sustained, a figure with no source	Revised after the independent review: 75 W is typical of leg pedalling, and sustained arm cranking is nearer 30 to 40 W. At 35 W each the three-person hoist takes about 111 s and the cycle about 3.7 min; with the lead and head sheaves (about 8 %) about 3.3 min at 50 W. A	**Plausible, to confirm at TRL 3**	Open: see below

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
					cited arm-crank figure and the timed trial decide; PR-004 is already not met on paper		
S9	Crank force for each of two people at the rated grab load	Biology	Ergonomics	Torque balance through the 4:1 chain (95 %) and two bearings (98 %), 250 mm crank, third rope layer; R5 limit 150 N	55 N <= 150 N	**Within known science**	SKG-CAL-001 D1
S10	The band brake holds the drum at the slip clutch's limit	Physics	Mechanics and statics	Band brake (capstan equation) T1/T2 = e^(mu theta), mu 0.35, 270 deg wrap; 6 kg on a 10.2:1 lever on the band's slack end, the anchored end tight	130 N m <= 314 N m	**Within known science**	SKG-CAL-001 F2
S10a	The band brake holds when the lining is wet or muddy	Physics	Mechanics and statics	Capstan equation with mu 0.20 to 0.25 for a wet lining	Added after the independent review: at mu 0.25 the brake gives about 168 N m and at 0.20 about 117 N m; with the slip clutch of decision O4 the line can apply at most 130 N m (186 N m with the shear pin), so a wet lining at mu 0.20 falls just below it but stays above the 103 N m the 25 m scraper string needs; keep the lining dry and covered, and measure the holding torque wet before use (TRL 4)	**Plausible, to confirm at TRL 3**	Open: see below

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S11	The weighted lever applies the brake when it is let go	Physics	Mechanics and statics	Lever kinematics: the link pin must move away from the band's live end as the weight falls	Kit 1.7 model: link pin 25 mm below the pivot on the well side, so the falling weight moved it toward the live end and slackened the band; fixed: pivot lowered to 67 mm, link pin on an upright arm 60 mm above it, link on the band's tangent; ratio about 10.3:1 (CAL uses 10.2)	**Within known science**	SKG-DEC-001 K2 (fix rule)
S12	8 mm polyester line has an adequate factor on the working pull	Engineering and technology	Standards practice	12 kN minimum breaking strength, 90 % at the splice; fibre rope slings use 5:1 to 7:1 (EN 1492-4); the reviewer finds catalogue values of 8 to 10 kN, which still give about 7.7	10.2 x >= 5 x	**Within known science**	SKG-CAL-001 E3
S13	Sheave pitch diameter suits the double-braid line	Engineering and technology	Standards practice	ANSI/ASSP A10.48: at least 6:1 for double-braid rope (Messer 2019)	16.2 D/d >= 6 D/d	**Within known science**	SKG-CAL-001 E4
S14	Fleet angle at the drum keeps the line winding evenly	Engineering and technology	Standards practice	Drum practice: fleet angle about 1.5 to 2 deg or less for a plain drum	1.91 deg <= 2 deg	**Within known science**	SKG-CAL-001 E2
S14a	The line spools evenly over three layers on the plain drum	Engineering and technology	Standards practice	Wire-rope guidance for a smooth drum is about 1.5 deg; multi-layer plain drums usually need grooving or a level wind	Added after the independent review: 1.91 deg is over 1.5 deg; the operator guides the line by hand on the third layer; to confirm in the timed trial	**Plausible, to confirm at TRL 3**	Open: see below

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S15	The drawbar does not buckle at the slip clutch's limit	Physics	Strength of materials	Euler buckling of a pinned 48.3 x 3.2 tube, E 200 GPa	1472 N <= 3.94e+04 N	**Within known science**	SKG-CAL-001 G2
S16	The capstan does not pitch forward at the slip clutch's limit	Physics	Mechanics and statics	Overturning couple of rope and drawbar against the capstan's weight about the front rail	118 N m <= 258 N m	**Within known science**	SKG-CAL-001 G3
S17	The 42.4 x 2.6 pole carries the scraper torque	Physics	Strength of materials	Torsion of a thin tube against shear yield 0.58 x 235 MPa	30 MPa <= 136 MPa	**Within known science**	SKG-CAL-001 H2
S18	The 25 m scraper string hangs inside the slip clutch's lowest slip on the first layer and the tripod rating	Physics	Mechanics and statics	String weight x 1.1 against 1,331 N (clutch, innermost layer, low end of the setting) and 1,472 N; sections come off as it rises	1295 N <= 1331 N	**Within known science**	SKG-CAL-001 H4
S19	Folded, the scraper toes pass the smallest ring bore	Mathematics	Numerical method	Four-bar linkage solved in the model; sleeve slides 300 mm on the cross pin	387 mm radius <= 400 mm radius	**Within known science**	SKG-CAL-001 H5
S20	The scraper arm's pivot eye closes round its 16 mm pin and clears the 70 mm hub	Engineering and technology	Manufacturing tolerances	An eye needs material round the hole; the hub's radius is the limit	Kit 1.7 model: pivot 45 mm from the pole axis, 10 mm outside the hub, so the 50 mm arm ended in an open half hole; fixed: pivot at 60 mm, an 18 mm radius eye clears the hub by 7 mm; toe radius unchanged at 615 mm, fold 387 mm	**Within known science**	SKG-DEC-001 K6 (design for construction)
S21	The 8-ring string sinks 3 m below the	Physics	Mechanics and statics	Self-weight against skin friction; skin	Sinks only while friction stays below 1.66 kPa (CAL	**Plausible, to**	Open: TRL 4 (SKG-DEC-001 O2,

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	water table under its own weight and eight saddles, without de-watering			friction = K sigma'v tan delta for soil in contact (K 0.5, delta 2/3 phi')	estimate 1 to 3 kPa for disturbed sand); soil in full contact at mid depth gives about 25 kPa, so sinking depends on the 40 mm overcut keeping the soil loose and off the ring. Field practice sinks rings this way with de-watering (Abbott; WaterAid); without it there is no published figure. Revised after the independent review: the central estimate (2 kPa) already exceeds the limit, the overcut is only at the toe and saturated sand closes onto the ring above it, and full contact would need about 270 kN against 18 kN; the claim fails on first order. Decided by Amish on 2026-10-25 (SKG-DEC-001 O2): bentonite slurry fed down a pipe into the overcut annulus is part of the method from the first trial, as in caisson practice and as water-well drillers use bentonite muds. No figure for the lubricated friction is claimed; the test pit must show it below the limit, so the row stays open	**confirm at TRL 3**	decided 2026-10-25)

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
					at TRL 4 and PR-001 is at risk		
S22	Sand does not boil into the rings during digging	Physics	Fluid mechanics	Quick condition needs an upward hydraulic gradient near 1 (Terzaghi); with no de-watering the water inside the rings stands level with the water outside	Gradient about zero while the crew does not bail; the grab removes about 21 L a bite, too little to draw the level down. Revised after the independent review: a fast hoist of a sealed grab can draw a transient gradient, and sand can run in under the edge (loss of ground) without boiling; hoist slowly off the bottom and keep the inside level with the outside; to confirm in the trial	**Plausible, to confirm at TRL 3**	Open: see below
S23	Four dip-tape readings resolve tilt finer than the 1 in 80 limit	Mathematics	Statistics and uncertainty	Two readings of +/-3 mm across 1,075 mm (root sum of squares, as CAL I5); resolves the top ring only, so ring flatness and joints add to it	0.0039 rad <= 0.0125 rad	**Within known science**	SKG-CAL-001 I5
S24	The grab hangs from its line about 8 to 9 degrees out of level	Physics	Mechanics and statics	A body hung from one point rests with its centre of gravity under that point	The line enters the guide tube 65 mm off the grab's axis, about 410 mm above the empty grab's centre of gravity; the grab settles level on the bottom once the line is slack, so the bite is unaffected; no change	**Within known science**	This file (observation from the detailed model)
M2	Cycle at 10 m with three	Mathematics	Orders of magnitude	Sum of lowering, crank change,			SKG-CAL-001 J2

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	people at the hoist is about 3.2 min			open, close, hoist (50 W each) and dumping	claimed 3.2 min, estimate 3.18 min (ratio 1.01)	**Within known science**	
M3	About 3.6 m3 of soil is removed for 3 m of sinking	Mathematics	Orders of magnitude	Cylinder to the toe radius	claimed 3.56 m3, estimate 3.56 m3 (ratio 1.00)	**Within known science**	SKG-CAL-001 I4
E1	The drum holds the line for a 30 m well	Engineering and technology	Power and link budgets	Turns per layer x pitch circumference over three layers against depth plus reeving and dead turns	38.7 m <= 41.7 m	**Within known science**	SKG-CAL-001 E1
E2	Proof load at 1.25 x the rated load is a conservative test for hand-powered lifting gear	Engineering and technology	Standards practice	Static test coefficients in lifting standards are typically 1.25 to 1.5; the applicable standard is to assess (SKG-REG-001)	Restated for Amish's decision O4 of 2026-10-25. The independent review found the old basis inconsistent (2 x the link's 122 kg release, 244 kg, exceeded the tripod's 225 kg proof). PR-007 now proofs at 1.25 x the rated load, the clutch's highest slip (150 kg): 188 kg, inside the HatchSide proof, with the clutch locked out for the test. 1.25 sits at the low end of static test coefficients in lifting standards; which standard applies is still to assess (SKG-REG-001)	**Plausible, to confirm at TRL 3**	Open: TRL 4 (SKG-DEC-001 O4, decided 2026-10-25)
E3	Every part fits: no two detailed parts	Engineering and technology	Manufacturing tolerances	Kit 1.9 D3 fidelity check (cad_check.py):	Passed for options A and B after the	**Within known science**	cad/ fidelity.json

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	overlap and every made part is drawable			interference over 5 mm3, bare primitives, fasteners, BOM lines	fixes of SKG-DEC-001 K6 to K20		
C1	No exhaust gas in the shaft: the method has no engine in or at the well	Chemistry	Environmental and atmospheric chemistry	Carbon monoxide comes from incomplete combustion; it collects in a shaft (Oxfam WASH guidance)	Hand capstan only; the gas check stays because the shaft can hold gas from the ground	**Within known science**	SKG-PRC-001 Safety
C2	The daily gas test shows the shaft air is safe	Chemistry	Analytical	CO2 is 1.5 times as dense as air and collects in wells; 40,000 ppm is immediately dangerous, yet it can take about 7 % CO2 to pull oxygen down to a 19.5 % alarm (Henderson 2010)	Kit 1.7 BOM: a four-gas detector (O2, CO, H2S, flammable), which does not read CO2; fixed: a portable 0 to 5 % CO2 safety monitor with alarms at 0.5 % and 3 % (BOM line 35, USD 399) used with it before work, after breaks and whenever anyone leans over the shaft. The independent review found that a monitor held at the collar reads only collar air; fixed: both instruments are lowered on a cord in a mesh bag and read at 2 m, at mid depth and 1 m above the water, held for at least the stated response time at each level, and a reading over range counts as a fail	**Within known science**	SKG-DEC-001 K5, K22 (fix rule)
C3	Steel tools and polyester	Chemistry	Inorganic	Painted mild steel, zinc and	Lead-free primer and paint on every		

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	rope do not contaminate the well water			polyester are common in wells; lead-based paint is the risk	part that enters the well (BOM line 34); the well is disinfected after the work as the Oxfam guidance describes	**Within known science**	SKG-BLD-001 section 7
B1	The assembled grab is never lifted by hand	Biology	Ergonomics	Team lifting: two people handle two thirds of the sum of their individual capacities, three people half (Network Rail, after HSE); 25 kg each	Kit 1.7 build plan: two people lift the assembled grab (52.5 kg) against a two-person capacity of 33 kg; fixed: the grab is assembled on the line under the tripod and moved only by the capstan; it travels in pieces of 25 kg or less	**Within known science**	SKG-DEC-001 K3 (fix rule)
B2	A full spoil tub is never lifted by hand	Biology	Ergonomics	Team lifting as B1; push and pull limits of about 200 N to start a load (ISO 11228-2, order of magnitude)	A tub with one bite holds about 50 kg, over the 33 kg two-person capacity, and the kit 1.7 design did not say how it is emptied; fixed: two people slide it off the closed doors on a board and tip it at the heap, about 247 N to start it	**Within known science**	SKG-DEC-001 K4 (fix rule)
B3	No single piece needs more than two people to carry	Biology	Ergonomics	Heaviest piece against R9's 25 kg and the 33 kg two-person team capacity	22.4 kg <= 25 kg	**Within known science**	SKG-CAL-001 K1
B4	Four people can turn the scraper for an hour at a time	Biology	Ergonomics	Push and pull force limits for repeated work (ISO 11228-2; EN 1005-3 for machinery) depend on	About 142 N for each of two people on the 1.4 m T-bar (SKG-CAL-001 H1), turning continuously, failed against sustained limits of about 100	**Plausible, to confirm at TRL 3**	Open: TRL 4 (SKG-DEC-001 O5, decided 2026-10-25)

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				frequency and duration	N or less after the independent review. Decided by Amish on 2026-10-25 (SKG-DEC-001 O5): a 2.0 m T-bar if the layout allows, else four people. The 2.0 m bar clears the HatchSide legs only below about 0.8 m above ground (SKG-CAL-001 H10), so four people turn it, two on each side: 71 to 84 N each (H9). The 150 N toe force is an estimate, so the timed scraping trial confirms it (TRL 4)		
B5	Nobody is in the well while SinkGrab works	Biology	Human physiology and biomechanics	Bad air and falling loads are the causes of death in well work; removing the person removes the exposure	Every task is worked from the surface; the doors cover the shaft whenever the grab is up	**Within known science**	SKG-PRC-001; PR-008
B6	Deepening does not leave the well contaminated	Biology	Microbiology	Digging stirs sediment and brings tools and hands to the water; disinfection after work is standard practice (Oxfam WASH)	SinkGrab makes no water-quality claim; the method ends with cleaning and disinfecting the well (out of scope, SKG-PRB-001)	**Within known science**	SKG-PRB-001 Out of scope
OB1	Option B: the bailer holds about 36 L above its valve seat	Mathematics	Units and dimensional analysis	Cylinder, 210 mm bore, seat to rim	claimed 36 L, estimate 36.2 L (ratio 0.99)	**Within known science**	SKG-OPT-001
OB2	Option B: the steady pull of	Physics	Mechanics and statics	Bailer, sand at 50 % fill and	1106 N <= 1127 N		SKG-OPT-001

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	the full bailer stays below the slip clutch's lowest slip			water in air x 1.11 sheave friction against the clutch's lowest slip (decision O4; the shear link it was sized on is gone); barrel 1.0 m		**Within known science**	(barrel 1.0 m)
OB3	Option B: spudding fills a dart-valve bailer in loose saturated sand	Physics	Fluid mechanics	Cable-tool and manual percussion drillers clear loose sand with dart-valve bailers worked up and down on the bottom	Fill assumed 50% (range 30 to 70 %), 18 L a run; no figure for a 220 mm bailer in a 1 m ring; to confirm in a test pit	**Plausible, to confirm at TRL 3**	Open: see below
OB4	Option B: pulling the line 0.5 m sideways over 2 m lifts the bailer about 0.24 m	Mathematics	Orders of magnitude	Geometry of a line deflected at mid span between two fixed points	claimed 0.24 m, estimate 0.236 m (ratio 1.02)	**Within known science**	SKG-OPT-001
OB5	Option B: two people can spud the bailer about 30 times a run	Biology	Ergonomics	Force to hold the line deflected: 2 T sin(theta); push and pull limits for repeated work (ISO 11228-2)	362 to 521 N in all as the bailer fills, 260 N each at the end, plus the jerk. Revised after the independent review: that exceeds the 150 N limit of PR-005 and the 200 N start limit of B2, so it fails by the document's own criteria; a jerk lever or a third person is the fix if Amish chooses option B (SKG-DEC-001 O1)	**Fails, fix proposed**	Awaiting Amish: SKG-DEC-001 O1
OB6	Option B: the rings sink as loose sand	Physics	Mechanics and statics	Cohesionless soil slumps to its angle of repose	Works in sand and gravel only; no way to cut under the	**Fails, fix proposed**	Awaiting Amish: SKG-DEC-001 O1

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	runs from under the cutting edge into the bailer's crater			under water; cohesive soil (clay, laterite) stands and does not run in	edge or to correct tilt except saddles. Revised after the independent review: the rings still have to overcome skin friction, as in S21, with no undercut, so it fails on first order like S21. The bentonite annulus Amish chose on 2026-10-25 (SKG-DEC-001 O2) needs the overcut that only option A cuts, so it does not close this row; it stays with the concept choice (O1)		
OB7	Option B: setting the bailer down on its dart stem opens the valve	Physics	Mechanics and statics	Bailer weight on the stem against the sand column on the 140 mm dart	302 N <= 464 N	**Within known science**	SKG-OPT-001

The fixes (SKG-DEC-001 K1 to K5 and K22)

1. **Shear link (S5, K1).** The 3:1 tackle's dead end hangs on a calibrated pin of about 1.9 mm that must release between 1.20 and 1.47 kN, so the line never loads the HatchSide tripod past its 150 kg rating. As drawn in kit 1.7 the rope's thimble hung on the bare pin across the 12 mm gap between the two link plates. A pin loaded across an open gap fails in bending long before it shears: about 0.15 kN here, so the grab would have opened on every lift. A hardened 16 mm sleeve now fills the gap and carries the thimble, with a hardened bush in each plate and clearances under 0.1 mm, so the pin is cut at the two faces of the sleeve as the calculation assumes. Pins are break-tested in a spare link, not loose. About USD 3. On 2026-10-25 Amish's decision O4 replaced the shear link with a load indicator that does not release (S5 restated).
2. **Brake that is on when let go (S11, K2).** The brake lever's link pin sat 25 mm below its pivot on the well side, so the falling weight pushed the link toward the band's live end and slackened it. The lever pivot is now 67 mm above the floor on a plate on the outside of the rail, and the link runs on the band's tangent to a pin on an upright arm 60 mm above the pivot. As the weight falls the pin moves away from the band, which tightens it; the ratio is about 10.3:1 against the 10.2:1 the calculation uses. About USD 3, with the new anchor eye and posts (K13).
3. **Assembled grab (B1, K3).** At 52.7 kg (52.5 kg before the load indicator) the assembled grab is beyond two people, whose team capacity is about two thirds of their combined 50 kg, and beyond three. The build plan now

assembles it on the line under the tripod and moves it only with the capstan; it travels in pieces of 25 kg or less. No cost.

4. **Spoil tub (B2, K4).** A tub holding one bite weighs about 50 kg and the kit 1.7 design did not say how it leaves the doors. Two people now slide it off the closed doors on a board to the ground and tip it at the heap, about 250 N to start it; it is never lifted full. No cost.
5. **Carbon dioxide (C2, K5).** Carbon dioxide is 1.5 times as dense as air and collects in wells. A four-gas detector reads oxygen, carbon monoxide, hydrogen sulphide and flammable gas but not carbon dioxide, and it can take about 7 % carbon dioxide to pull oxygen down to the usual 19.5 % alarm, while 4 % is immediately dangerous (Henderson 2010). A portable 0 to 5 % carbon dioxide safety monitor (BOM line 35, USD 399) is now used beside the four-gas detector before work, after breaks and whenever anyone leans over the shaft; new requirement PR-012 asks for this air test. Nobody enters the well in any case.
6. **Gas read at depth (C2, K22).** The independent review found that an instrument held at the collar reads only the collar's air, while carbon dioxide pools at depth in still air. Both instruments are now lowered on a cord in a mesh bag and read at 2 m below the collar, at mid depth and 1 m above the water, each held for at least its stated response time; a reading over range counts as a fail. No cost.

Rows decided by Amish on 2026-10-25

- **S21, sinking without de-watering (PR-001, SKG-DEC-001 O2).** The string sinks to 3 m only while skin friction stays below 1.66 kPa; the central unaided estimate (2 kPa) already stops it, and soil in full contact gives about 25 kPa. Decided: bentonite slurry fed down a pipe into the 40 mm overcut annulus is part of the method from the first trial, tried first in the test pit. No figure for the lubricated friction is claimed, so the row is plausible, to confirm at TRL 4; PR-001 is at risk.
- **S4 to S7 and E2, overload device and proof basis (PR-007, O4).** The shear link's window left 3 N to the rating, sheave friction and breakout suction made it trip or overload, and the crank pin ran above shear yield; twice the rated load exceeded the tripod's proof. Decided: a friction slip clutch on the crank shaft, set on CalRig to slip between 1,127 N (third layer) and 1,472 N (first layer) at the capstan, and a peak-reading load indicator at the dead end; proof at 1.25 times the 150 kg rated load, 188 kg. The steady full-grab pull at the capstan is 1,068 N, a margin of 1.06 on the third layer on a ±5 % setting, so S6, S7 and E2 are plausible, to confirm on CalRig; S4 passes on paper.
- **B4, turning the scraper (O5).** 142 N each for two people against about 100 N. Decided: a 2.0 m T-bar if the layout allowed it, else four people; the 2.0 m bar clears the tripod legs only below about 0.8 m above ground, so four people turn the 1.4 m bar, 71 to 84 N each; plausible, to confirm in the timed trial.

Rows that fail, with a fix proposed

- **OB6, sinking with the bailer (option B, O1).** Option B has the same skin friction as S21 and no undercut, so the bentonite annulus of O2, which needs the overcut, does not close it. It waits on the concept choice.
- **OB5, jerking the bailer (option B, O1).** Up to 260 N each at the end of a stroke, over the 150 N of PR-005. Fixed only if option B is chosen: a jerk lever or a third person.

Rows that stay open

- **S2, closing force in sand.** Through the 3:1 tackle each lip closes with about 349 N against about 270 N of passive drag at full bite depth in loose saturated sand. The review found that closing starts with the grab empty,

that the three sheaves lose 10 to 15 %, and that whether 1.06 times the line pull acts on each lip or on both is not settled; the margin lies between about 0.9 and 1.8. Closed by the test-pit bite trial (TRL 4) in each soil; PR-003 is at risk.

- **S3, opening by head weight.** The 14.3 kg head opens the shells when the line is slacked; whether sand lying on the lips resists it is a trial question.
- **S8, hoisting power.** The 75 W sustained figure has no source and suits leg pedalling; arm cranking is nearer 30 to 40 W, which puts the cycle near 3.7 min. A cited arm-crank figure and the timed trial decide.
- **S10a, wet brake.** At a friction coefficient of 0.25 the brake holds about 168 N m, over the 130 N m the slip clutch now lets the line apply (186 N m with the crank pin); at 0.20 it holds about 117 N m, just under, but over the 103 N m the scraper string needs; measured wet before use.
- **S14a, spooling.** The 1.91 degree fleet angle is over the 1.5 degrees of wire-rope guidance for a plain drum; the operator guides the third layer by hand.
- **S22, quick condition and loss of ground.** A fast hoist of a sealed grab can draw a transient gradient, and sand can run in under the edge without boiling; hoist slowly off the bottom and keep the inside level with the outside.
- **S5, S6, S7, E2, S21 and B4.** Restated for the decisions of 2026-10-25 (above); closed by CalRig, the test pit and the timed scraping trial.
- **OB3 (option B).** Bailer fill by spudding.

Fields not relevant

None. Every field applies: physics (statics, strength of materials, fluid mechanics of the flooded well), chemistry (well gases, coatings), biology (ergonomics, manual handling, physiology of bad air, water hygiene), mathematics (volumes, cycle and tilt arithmetic) and engineering (rope and sheave practice, fits in the detailed model, proof testing). Electromagnetism, optics, acoustics and nuclear branches have no claims: SinkGrab has no electrical or optical parts.

Concept options

- **Option A, rope grab and under-curb scraper:** mechanism within known science; after Amish's decisions of 2026-10-25 nothing fails; S2, S3, S5, S6, S7, S8, S10a, S14a, S21, S22, E2 and B4 to confirm by trial.
- **Option B, dart-valve bailer on a jerk line:** within known science (OB1, OB2, OB4, OB7), with OB3 (fill) plausible and OB5 (jerk effort) and OB6 (sinking, as S21) failing with a fix proposed. At 1.2 m long the full bailer reached 1,260 N, over the shear link's lowest release; the option is modelled with a 1.0 m barrel (1,096 N), whose steady pull stays under the slip clutch's lowest slip that now replaces the link (OB2). It does not undercut, so it works only where the soil runs in (sand and gravel), and it corrects tilt only with saddles (SKG-OPT-001).

Independent review

On 2026-10-25 an independent reviewer, a second Claude agent run headless (`claude -p`) that did not see this session's reasoning, was given only this STEM check (as first written), the calculation note SKG-CAL-001, the product requirements SKG-PRS-001 and `docs/04-calcs/sizing.py`, and asked to refute every "Within known science" verdict. It reproduced the arithmetic of nearly every row and found that several verdicts rested on assumptions the check had not stressed. What it found and what was done:

Table 2. Independent review findings and responses.

Finding	Rows	Response
Ring sinking is refuted by the note's own numbers: the central friction estimate (2 kPa) exceeds the 1.66 kPa limit, full contact (about 25 kPa) needs about 270 kN against 18 kN, and saturated sand closes onto the ring above the overcut	S21, OB6, PR-001	Agreed. Both rows now fail with a fix proposed; PR-001 is not met on paper; a lubricated annulus or jacking is posed to Amish (SKG-DEC-001 O2), because it changes the method
The proof basis is inconsistent: twice the rated line load (244 kg) exceeds the tripod's 225 kg proof, a 196 kg test exceeds both overload devices' releases, and E2 printed 193 kg against CAL's 196 kg	E2, PR-007	Agreed. E2 fails with a fix proposed; the figure now quotes CAL's 196 kg; PR-007 is at risk; posed to Amish (O4)
The shear link passes by 3 N on a back-solved pin no stock wire meets; sheave friction puts the line pull at release over the rating; breakout suction can trip it in use	S7, S4	Agreed. S7 fails with a fix proposed (O4); S4's law now says suction and friction are not included
The crank pin runs at about 70 % of its release torque in normal hoisting, above shear yield, and the 240 MPa basis is low for S235	S6, S18	Agreed. S6 fails with a fix proposed (O4: a calibrated slip clutch is recommended). S18 keeps its verdict for the string's weight but inherits O4
The closing force has a factor-of-two ambiguity, uses the full rather than the empty grab, and leaves out sheave friction	S1, S2, PR-003	Agreed in part: S1's ratio stands; S2 is now plausible with the margin stated as about 0.9 to 1.8; PR-003 is at risk until the bite trial
The 75 W sustained figure has no source; arm cranking is nearer 30 to 40 W	S8, M2, PR-004	Agreed. S8 is now plausible; at 35 W each the cycle is about 3.7 min. PR-004 was already not met on paper
The brake's slack side was not stated, and a wet lining holds less	S10, S11	S10's law now states the weight acts on the slack end; a new row S10a (wet lining) is plausible. The reviewer's note that `sizing.py` uses lever constants from before K2 was checked: the script's 10.2:1 and the model's 10.3:1 differ by 1 %, recorded in S11
The rope's 12 kN is optimistic, the factor 5 uncited, and the fleet angle over the 1.5 degrees of wire-rope guidance	S12, S14	S12's law now cites EN 1492-4 practice and the reviewer's 8 to 10 kN (factor about 7.7, still above 5); new row S14a (spooling) is plausible. MBS to be read from the supplier's datasheet when bought
The quick-condition row ignores transient drawdown and loss of ground	S22	Agreed. S22 is now plausible, with slow hoisting off the bottom and level control
Tilt resolution differs between CAL (root sum of squares) and this check	S23	Fixed: S23 now uses the root sum of squares, as CAL I5; the row now says it resolves the top ring only
The scraper's 142 N each exceeds sustained push limits, and option B's 260 N each exceeds the document's own limits	B4, OB5	Agreed. Both fail with a fix proposed: B4 to Amish (O5, a longer T-bar or four people); OB5 only if option B is chosen (O1)
A gas monitor held at the collar reads only collar air	C2	Fixed under the standing instruction (K22): both instruments are lowered and read at three depths
Manual handling: 25 kg is the best-zone guideline, and the 45 s dump allowance is short	B1 to B3, M2	Noted. B3's claim is carriage by two people, which the 22.4 kg frame meets; the dump time is part of the timed trial

Finding	Rows	Response
The count of first-check failures omitted S20	Table 1 caption	Fixed: six rows failed on first check (S20 fixed as design for construction, K6)
Missing claims: dropped-grab shock on the recovery line, breakout suction, friction through the tackle, pole couplings and string buckling, tripod overturning under a snag, ring joints during sinking, rope abrasion, and the time for the whole job	New	Listed in SKG-CAL-001 section O for the next revision; not added as rows in this session because each needs a calculation, not a screen

The reviewer's overall judgement was that the "open" verdict with none failing was too generous. After the responses above the check was still open, with seven rows failing with a fix proposed, all awaiting Amish (five of them decided on 2026-10-25, above), and none failing in a way that stops the digging principle itself. The reviewer could not see `cad/src/model.py` or HatchSide's calculation note HTS-CAL-001, so the tripod's 150 kg rating and the model-derived values were taken as given.

Sources

- Abbott, Hand Dug Wells: Choice of Technology and Construction Manual; WaterAid technology brief on hand-dug wells; Oxfam WASH, "Repairing, Cleaning and Disinfecting Hand Dug Wells": see SKG-PRB-001.
- Henderson, Bob. "Measuring Carbon Dioxide Levels." *Safety+Health*, September 1, 2010. https://www.safetyandhealthmagazine.com/articles/measuring-carbon-dioxide-levels-2 (checked 2026-10-25).
- Messer, Caleb. "Rope and Rigging Compatibility, Part Two of Two." *Inside Towers*, September 6, 2019, citing ANSI/ASSP A10.48. https://insidetowers.com/cell-tower-news-rope-and-rigging-compatibility-part-two-of-two/ (checked 2026-10-25).
- Network Rail. "Lifting and Handling in Teams," after HSE guidance. https://safety.networkrail.co.uk/safety/manual-handling/principles-of-manual-handling/lifting-and-handling-in-teams/ (checked 2026-10-25).
- CO2Meter. "Portable 5% CO2 Safety Detector." https://co2meter.com/products/personal-co2-monitor, listed at USD 399 (checked 2026-10-25).
- Terzaghi, Karl, Ralph B. Peck and Gholamreza Mesri. *Soil Mechanics in Engineering Practice*. 3rd ed. New York: Wiley, 1996 (quick condition, earth pressure, skin friction K sigma'v tan delta).
- ISO 11228-2 (manual handling, pushing and pulling) and EN 1005-3 (force limits for machinery operation), as cited; values are orders of magnitude here and the assessments are still to do.
- SKG-CAL-001 (`docs/04-calcs/01-sizing.md`) for every design figure.